

Quiz 3

Student Name: _____

Points: 20

A Number: _____

Duration: 30 Minutes

1. Consider the following code for the Generator:

```

class Generator(nn.Module):
    def __init__(self, in=28, height=28, width=28):
        super(Generator, self).__init__()
        self.in = in
        self.height, self.width = height, width

        self.model = nn.Sequential(
            nn.Linear(in, 256),
            nn.LeakyReLU(0.2),
            nn.Linear(256, 512),
            nn.LeakyReLU(0.2),
            nn.Linear(512, 1024),
            nn.LeakyReLU(0.2),
            nn.Linear(1024, width*height),
            nn.Tanh()
        )

    def forward(self, z):
        img = self.model(z)
        return img.view(img.size(0), 1,
                        self.height, self.width)

```

Based on the above code, what's the dimension of the noise that Generator uses in the default settings for mapping it to a possible image?

(a) 256

(b) 28 ✓

(c) 512

(d) 1024

2. Consider the Bayesian framework:

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})}$$

with $p(\mathcal{D}) = \int p(\mathcal{D}|\theta)p(\theta)d\theta$.

Here, θ is a parameter or latent representation. $\mathcal{D}$ is dataset.

Out of all distributions, which is the most difficult to model?

(a) $p(\theta|\mathcal{D})$

(b) $p(\mathcal{D})$ ✓

(c) $p(\theta)$

(d) $p(\mathcal{D}|\theta)$

3. How many Neural networks are present in Vanilla GAN:

(a) 3

(b) 1

(c) 2 ✓

(d) 4

4. Explain cycle-consistency as you learnt in Cycle GAN.

5. Explain what are some issues encountered in GAN training.

6. If we have an input matrix as

$$\begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$$

and we are looking to apply fractionally-strided convolution with a stride of 2, then how does the input change before applying convolution using a kernel K .

(a)

$$\begin{bmatrix} 3 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \\ 7 & 0 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \checkmark$$

(b)

$$\begin{bmatrix} 3 & 0 & 5 \\ 0 & 0 & 0 \\ 7 & 0 & 9 \end{bmatrix}$$

(c)

$$[3 \ 5 \ 7 \ 9]$$

(d)

$$\begin{bmatrix} 3 \\ 5 \\ 7 \\ 9 \end{bmatrix}$$

7. What does AutoEncoder do?

- (a) It creates a compressed representation of dataset. ✓
- (b) It adds noise to dataset.
- (c) It can generate sythetic samples never seen before based on a training set.
- (d) It enhances image resolution.

8. Which one is the correct formula for KL Divergence **from** distribution q **to** distribution p ?

(a)

$$KL(q(\theta)||p(\theta|\mathcal{D})) = \int p(\theta) \log \frac{p(\theta|\mathcal{D})}{q(\theta)} d\theta$$

(b)

$$KL(q(\theta)||p(\theta|\mathcal{D})) = \int p(\theta) \log \frac{q(\theta)}{p(\theta|\mathcal{D})} d\theta$$

(c)

$$KL(q(\theta)||p(\theta|\mathcal{D})) = \int q(\theta) \log \frac{q(\theta)}{p(\theta|\mathcal{D})} d\theta \quad \checkmark$$

(d)

$$KL(q(\theta)||p(\theta|\mathcal{D})) = \int p(\theta) \log p(\theta|\mathcal{D})$$

9. Mean-field variational family assumes that

- (a) latent variables are independent of each other in the approximate posterior. 
- (b) latent variables are orthogonal to each other in the approximate posterior.
- (c) latent variables are linearly dependent on each other in the approximate posterior.
- (d) latent variables have stochastic.

10. What do you understand by prior in Bayesian framework?

11. The ELBO is equal to the negative KL divergence up to an additive constant.

- (a) True 
- (b) False

12. In the context of Variational Autoencoder which one is not true, assuming x as the input, z as the latent variable, and q as tractable approximation of the original distribution p ?

- (a) $q_w(z|x)$ maps x to a probability distribution over a latent space.
- (b) $p(z|x)$ is the intractable posterior, that is, it is very difficult to model.
- (c) $p(x)$ is known and you can always write easily. 
- (d) $q_w(z|x)$ is the probabilistic encoder.

13. Consider a random variable

$$\varepsilon \sim \mathcal{N}(0, \mathbf{I})$$

then the new random variable

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \varepsilon,$$

for constant $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ is a

- (a) A Gaussian distribution. ✓
- (b) A Uniform distribution
- (c) An intractable distribution
- (d) A direc-delta distribution

14. Consider a random variable

$$\varepsilon \sim \mathcal{N}(0, \mathbf{I})$$

Imagine that $\mathbf{x}_0$ is the initial data distribution (the original data distribution).

Then, in the forward diffusion process, we have

$$\mathbf{x}_{t+1} = \sqrt{1 - \beta_t} \mathbf{x}_t + \sqrt{\beta_t} \varepsilon$$

Here, the recursive relationship collapses into a single step relative to $\mathbf{x}_0$:

$$\mathbf{x}_{t+1} = \sqrt{\bar{\alpha}_{t+1}} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t+1}} \varepsilon$$

Choose the correct option that provides relationship between α_i and β_i

- (a) $\alpha_i = \beta_i$, and $\bar{\alpha}_{t+1} = \sum_{i=0}^t \alpha_i$
- (b) $\alpha_i = 1 + \beta_i$, and $\bar{\alpha}_{t+1} = \sum_{i=0}^t \alpha_i$
- (c) $\alpha_i = 1 - \beta_i$, and $\bar{\alpha}_{t+1} = \prod_{i=0}^t \alpha_i$ ✓
- (d) $\alpha_i = 1 + \beta_i$, and $\bar{\alpha}_{t+1} = \prod_{i=0}^t \beta_i$

15. Referring to the previous problem, suppose somebody told you that initial data $\mathbf{x}_0$ has distribution $\mathcal{N}(m, s)$ where s is the variance and m is the mean.

In such a case, what is the mean/expectation of $\mathbf{x}_{t+1}$ in the forward diffusion process (Hint: use the property of expectations):

- (a) $\mathbb{E}[\mathbf{x}_{t+1}] = \sqrt{\bar{\alpha}_{t+1}} m$ ✓
- (b) $\mathbb{E}[\mathbf{x}_{t+1}] = \sqrt{\alpha_{t+1}} m$
- (c) $\mathbb{E}[\mathbf{x}_{t+1}] = \sqrt{\beta_{t+1}} m$
- (d) $\mathbb{E}[\mathbf{x}_{t+1}] = \sqrt{1 - \beta_{t+1}} m$

16. An autonomous equation doesn't depend explicitly on

- (a) Time variable t ✓
- (b) Input variable x
- (c) Output variable y

(d) Derivative of the output variable $\frac{dy}{dt}$.

17. Which one is true for Brownian motion:

- (a) Any sample at time t is Gaussian distributed.
- (b) Increment between two consecutive time is Gaussian distributed. ✓
- (c) Increments are dependent on the previous increment.
- (d) In Brownian motion, the variance of the increment decreases with time.

18. When does SDE collapses to become ODE?

- (a) When the diffusion coefficient $\sigma_t = 0$. ✓
- (b) When the diffusion coefficient $\sigma_t = \sigma$, a constant.
- (c) When the derivative of diffusion coefficient $\frac{d\sigma_t}{dt} = 0$.
- (d) When the derivative of diffusion coefficient σ_t cannot be determined.

19. In ODE interpretation of Diffusion Model, ODE is defined via a time-dependent vector field $u : [0, 1] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$. u_t is the target vector field at time t . If we are to approximate u_t using a neural network u_t^θ , what loss function we are interested to train u_t^θ ?

- (a) $\mathcal{L}_{FM}(\theta) = \mathbb{E}_{t, X_t} \left[||u_t^\theta(X_t) - u_t(X_t)||^2 \right]$ ✓
- (b) $\mathcal{L}_{FM}(\theta) = \mathbb{E}_{t, X_t} \left[||u_t^\theta(X_t) + u_t(X_t)||^2 \right]$
- (c) $\mathcal{L}_{FM}(\theta) = \mathbb{E}_{t, X_t} \left[||u_t^\theta(X_t | X_{t-1}) - u_t(X_t)||^2 \right]$
- (d) $\mathcal{L}_{FM}(\theta) = \mathbb{E}_{t, X_t} \left[||u_t^\theta(X_t | X_{t+1}) - u_t(X_t)||^2 \right]$